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Drivers of Nutritional Transition in Kenyan Children: Examining the Relative Contribution of Women's Education, Economic Development, and Sanitation Infrastructure to Reductions in Childhood Stunting (1993-2022)

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DECLARATION

I, Lorna Kendi Etere, declare that this research proposal, titled **"Drivers of Nutritional Transition in Kenyan Children: Examining the Relative Contribution of Women's Education, Economic Development, and Sanitation Infrastructure to Reductions in Childhood Stunting (1993–2022)"**, is my original work. It has not been submitted for any degree or examination at any other university. All sources of information have been duly acknowledged through citations and references.

Signature: ___________________________

Date: ___________________________

Student Name: Lorna Kendi Etere

Student Registration: AST/0222/23

Supervisor:

DEDICATION

To the children of Kenya—past, present, and future—may each generation grow stronger than the last.

And to my family, for their unwavering support and patience throughout this journey.

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my supervisor, for their invaluable guidance, constructive feedback, and encouragement throughout the development of this research proposal. Their expertise and patience have been instrumental in shaping this work.

I am also grateful to the Demographic and Health Surveys (DHS) Program for making high-quality data publicly available, enabling research such as this to contribute to evidence-based policy and practice.

Special thanks to my family for their understanding and support during the long hours required to complete this work. Their belief in me has been a constant source of motivation.

Finally, I acknowledge the broader community of researchers and practitioners working to improve child nutrition in Kenya and across Africa. This work is part of a larger effort to which I am proud to contribute.

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ABSTRACT

Background: Child malnutrition remains a critical public health challenge globally, with long-term consequences for individual development and national economic productivity. Kenya has made remarkable progress in reducing child stunting over the past three decades, yet the relative contribution of different drivers remains underexplored. **Objective:** This study examined trends and determinants of child malnutrition in Kenya between 1993 and 2022, focusing on the relative contributions of women's education, household wealth, and environmental factors to stunting reduction. **Methods:** Data from seven Kenya Demographic and Health Survey rounds (1993–2022) were analyzed. Trends in stunting, wasting, and overweight were documented alongside changes in women's secondary education, household electrification, television ownership, and water and sanitation access. Pearson correlation and simple linear regression were used to assess associations with stunting. **Results:** Stunting declined by 22.3 percentage points, from 39.9% to 17.6%. Women's secondary education more than doubled (24.5% to 58.2%), and electricity access increased nearly sixfold (8.8% to 51.1%). Both were near-perfectly correlated with stunting reduction ($r = -0.988$ and -0.986 respectively) and explained over 96% of stunting variation in regression models (both $p < 0.05$). Improved sanitation, despite quintupling in coverage, showed a weaker, non-significant association ($p = 0.157$). **Conclusion:** Kenya's stunting reduction has been driven primarily by investments in women's education and broad-based economic development. Sanitation improvements alone appear insufficient without accompanying behavior change. Findings underscore the need for integrated policies addressing education, economic inclusion, and hygiene behavior alongside infrastructure. **Keywords:** child malnutrition, stunting, Kenya, women's education, household electrification, sanitation, DHS data

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Childhood malnutrition remains one of the most persistent and damaging public health challenges globally, with profound consequences for both individual well-being and national development. Malnutrition manifests in multiple forms, each reflecting distinct biological processes and temporal dynamics. **Stunting**, or chronic undernutrition, occurs when a child is too short for their age, indicating prolonged nutritional deprivation and recurrent illness. **Wasting** represents acute undernutrition, characterized by a child being too thin for their height due to recent and severe weight loss, often precipitated by food shortages or disease. These indicators are measured separately because they capture different time scales and etiologies: stunting reflects cumulative historical deprivation, while wasting signals immediate nutritional crisis (WHO, 2023).

The consequences of childhood malnutrition are severe and lifelong. From an individual child's perspective, malnutrition damages brain development, weakens immune function, and limits physical growth—often permanently. A stunted or wasted child is more likely to struggle academically, earn less as an adult, and face chronic health problems throughout life (Victora et al., 2021). As one observer notes, malnutrition "quietly steals potential before life even begins." From a national development perspective, widespread child malnutrition systematically weakens the future workforce, suppresses economic productivity, and elevates healthcare expenditures. A nation with high stunting rates sacrifices human capital accumulation and constrains its long-term economic growth trajectory (World Bank, 2022). For a country like Kenya aspiring to sustainable prosperity, undernourishing the next generation represents an unacceptable developmental cost.

Globally, the burden of malnutrition remains substantial despite decades of intervention. According to the most recent joint estimates from UNICEF, WHO, and the World Bank (2023), approximately 148 million children under five years of age are stunted, 45 million are wasted, and 37 million are overweight—underscoring that malnutrition encompasses both undernutrition and overnutrition. Nearly one in five children worldwide is stunted, and wasting contributes to an estimated 45% of all deaths among children under five (Black et al., 2020).

Africa bears a disproportionate share of this burden, accounting for roughly one-third of the world's stunted children. The continent has experienced slower reductions in stunting compared to Asia, and in many sub-Saharan African countries, stunting rates persist above 30%—a threshold the World Health Organization classifies as a "high" public health concern (WHO, 2023). Within East Africa, Kenya has made measurable progress, yet challenges remain. Data from recent Demographic and Health Surveys indicate that national stunting prevalence has declined from approximately 26% to 18% over the past decade, while wasting has stabilized around 4–5% (KNBS & ICF, 2023). However, these national averages mask considerable sub-national variation: arid and semi-arid lands (ASAL) counties consistently report stunting rates exceeding 30%, and wasting spikes during drought periods, highlighting persistent vulnerabilities (Ministry of Health, Kenya, 2022).

1.2 Statement of the Problem

Kenya has achieved remarkable progress in reducing child stunting over the past three decades. National prevalence declined from nearly 40% in 1993 to 17.6% in 2022—a reduction that has improved the lives of millions of children and strengthened the country's human capital base. This success warrants careful examination. Understanding what drove these improvements is essential for sustaining momentum and for reaching the children in marginalized regions who have not yet benefited equally.

Existing literature has identified numerous correlates of child malnutrition: maternal education, household wealth, water and sanitation access, and health-seeking behavior, among others (Black et al., 2020; Victora et al., 2021). However, most studies examine these factors in isolation or within single time points. What remains less understood is the **relative contribution** of different factors over time. When multiple variables improve simultaneously—as they did in Kenya between 1993 and 2022—which ones matter most? Is it education, which empowers mothers with knowledge and agency? Is it economic development, which brings electricity and information into homes? Or is it infrastructure, such as improved sanitation, which reduces disease exposure?

The distinction is not academic. If women's education is the primary driver, then investments in girls' secondary schooling should be prioritized. If electricity and household wealth matter more, then poverty reduction and infrastructure development take precedence. If sanitation alone is insufficient without accompanying behavior change, then past interventions may have been misdirected.

Preliminary analysis of Kenyan Demographic and Health Survey data suggests that women's secondary education and household electrification are exceptionally strong predictors of stunting reduction, explaining over 96% of the variation between 1993 and 2022. In contrast, improved sanitation—despite expanding from 13.6% to 66.6% over the same period—shows no statistically significant independent association with stunting. This finding challenges conventional assumptions and demands rigorous investigation.

This study therefore seeks to systematically examine the trends and determinants of child malnutrition in Kenya over three decades, with particular attention to the relative contributions of maternal education, household wealth, and environmental factors. By doing so, it aims to provide evidence that can inform more targeted and effective nutrition policies in Kenya and similar contexts across sub-Saharan Africa.

1.3 Research Objectives

General Objective

To examine the trends and determinants of child malnutrition in Kenya between 1993 and 2022, with particular focus on the relative contributions of maternal education, household wealth, and environmental factors to stunting reduction.

Specific Objectives

1. To analyze trends in stunting, wasting, and overweight among Kenyan children under five years of age from 1993 to 2022.
2. To assess changes in key socio-demographic indicators—including maternal education, household electrification, television ownership, and improved sanitation access—over the same period.
3. To determine the strength of association between each socio-demographic indicator and childhood stunting using correlation and regression analysis.
4. To compare the relative predictive power of maternal education, household wealth proxies, and environmental factors in explaining stunting reduction.
5. To identify which factors have been most strongly associated with Kenya's progress in reducing child stunting and discuss implications for policy.

1.4 Significance of the Study

This study holds significance for multiple audiences:

For policymakers in Kenya, the findings provide evidence to guide resource allocation. If women's secondary education emerges as the dominant predictor of stunting reduction—as preliminary analysis suggests—then investments in girls' education are not merely social policy but direct nutrition interventions. If household electrification and television access are equally powerful, then poverty reduction and infrastructure development warrant parallel priority. The counterintuitive finding regarding sanitation—that infrastructure expansion alone showed no significant independent association—challenges conventional wisdom and suggests that future sanitation programming must integrate hygiene education and behavior change components to achieve nutritional impact.

For public health practitioners and program designers, this study offers insight into which interventions have historically worked in Kenya and which have not. Understanding the relative contribution of different factors over a 30-year period provides empirical grounding for theory of change models and helps prioritize limited resources toward the most effective strategies.

For researchers in nutrition, epidemiology, and development studies, this study contributes to a growing literature on the social determinants of child health. By analyzing multiple time points and comparing the predictive power of diverse indicators, the study moves beyond simple correlation to assess relative contribution—a methodological approach that can inform future research design.

For international development partners—including UNICEF, WHO, the World Bank, and bilateral donors—Kenya's experience offers lessons for other countries in sub-Saharan Africa facing similar nutrition challenges. Understanding what drove Kenya's success can inform regional strategies and investment priorities.

For Kenyan communities and families, the ultimate significance of this research lies in its potential to inform policies that reach the children still left behind. The children in arid and semi-arid counties who continue to experience stunting rates above 30% are the ones this research ultimately serves.

1.5 Scope of the Study

Temporal Scope

This study covers the period from 1993 to 2022, encompassing seven Demographic and Health Survey (DHS) rounds conducted in Kenya during this timeframe. These years were selected because they represent the full span of available nationally representative data on child anthropometry and socio-demographic characteristics, allowing for analysis of long-term trends.

Geographic Scope

The study focuses on Kenya at the national level. While DHS data include regional identifiers, this analysis examines aggregate national trends to establish overall patterns. Future research may extend this work to sub-national analysis.

Population Scope

The study focuses on children under five years of age, the standard age group for malnutrition surveillance. Nutritional indicators examined include stunting (height-for-age), wasting (weight-for-height), and overweight. Maternal education, household assets, water and sanitation access are examined as predictors.

Content Scope

The study examines the relationship between child malnutrition and a defined set of socio-demographic indicators: maternal education, household electrification, television ownership, refrigerator ownership, improved water source, and improved sanitation. These indicators were selected based on:

- Availability across all survey rounds

- Theoretical relevance from prior literature
- Demonstrated variation over the study period

A note on data availability: Maternal literacy data are only available from 2003 onward. For this reason, literacy is included in descriptive trend analysis but excluded from the full longitudinal regression modeling, which requires consistent measurement across all time points. This limitation is acknowledged and addressed methodologically.

Factors such as dietary diversity, food security, micronutrient supplementation, and health service utilization—while undoubtedly important—are not included due to inconsistent measurement across survey rounds. The study therefore focuses on structural and socio-demographic determinants rather than proximate nutritional inputs.

1.6 Limitations of the Study

This study, while rigorous within its defined scope, is subject to several limitations that should be considered when interpreting findings.

Data Limitations
First, the analysis relies on secondary data from Demographic and Health Surveys, which are cross-sectional in nature. While we examine trends across multiple survey rounds, the data at each time point represent different samples of children. This precludes true longitudinal analysis of individual children over time and limits our ability to establish causal relationships. Associations identified should be interpreted as correlations rather than causal effects.

Second, the study is restricted to indicators measured consistently across all survey rounds. This necessitated excluding potentially important variables such as dietary diversity, food security, and health service utilization, which were not available for earlier survey years. The focus on structural determinants—while valuable—provides an incomplete picture of the multifaceted causes of malnutrition.

Third, maternal literacy data were only available from 2003 onward, limiting their inclusion in the full longitudinal analysis. While descriptive trends are presented, literacy could not be incorporated into the primary regression models alongside variables measured across all seven time points.

Measurement Limitations
All indicators are based on self-report or household observation and are subject to potential measurement error. Wealth proxies (electricity, television, refrigerator), while commonly used in DHS analysis, are imperfect measures of household economic status. Similarly, improved water source and sanitation facility definitions follow standard DHS classifications but may not capture consistency of access or quality of service.

Analytical Limitations
The regression analysis examines each predictor individually due to the limited number of time points (n=6 for the full analysis). While this approach effectively identifies the strength of bivariate relationships, it does not permit multivariate modeling that could assess the independent contribution of each factor while controlling for others. The strong correlations between predictors (e.g., education and electricity are themselves correlated) mean that their relative contributions cannot be fully disentangled with the available degrees of freedom.

Generalizability
Findings are specific to Kenya and may not generalize to other contexts with different cultural, economic, or epidemiological profiles. However, the analytical approach may be transferable, and Kenya's experience may offer lessons for similar settings.

1.7 Organization of the Study

This research proposal is structured into five chapters, each addressing a distinct component of the study.

Chapter One: Introduction has provided the background to the study, statement of the problem, research objectives, significance, scope, and limitations. It has established the rationale for examining trends and determinants of child malnutrition in Kenya and identified the specific gaps this research addresses.

Chapter Two: Literature Review will critically examine existing theoretical and empirical literature on child malnutrition. It will review global, African, and Kenyan studies, identify research gaps, and situate the current study within the broader academic discourse.

Chapter Three: Research Methodology will describe the research design, target population, data sources, and analytical techniques employed. It will detail the process of data extraction, cleaning, merging, and statistical analysis, as well as ethical considerations.

Chapter Four: Data Analysis, Interpretation, and Discussion will present the findings of the study, organized according to the research objectives. It will include descriptive trend analysis, correlation matrices, regression results, and visual representations of key relationships, followed by interpretation and discussion of findings in light of existing literature.

Chapter Five: Summary of Findings, Conclusions, and Recommendations will synthesize the main findings, draw conclusions, and offer recommendations for policy, practice, and future research.

References and appendices will follow the main text.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews existing literature on child malnutrition, with particular attention to its prevalence, determinants, and trends. The review progresses from global patterns to the African context and finally to Kenya, establishing the empirical and theoretical foundation for the present study. It concludes by identifying research gaps and explaining how this study contributes to filling them.

2.2 Global Studies on Child Malnutrition

Globally, child malnutrition remains a pressing public health challenge despite decades of intervention. According to joint estimates by UNICEF, the World Health Organization, and the World Bank (2023), approximately 148 million children under five years of age are stunted, 45 million are wasted, and 37 million are overweight. These figures represent not merely statistical aggregates but millions of individual children whose physical and cognitive development has been compromised.

The determinants of child malnutrition are multifaceted and interconnected. Black et al. (2020), in a comprehensive series published in *The Lancet*, identified maternal education, household socioeconomic status, dietary diversity, water and sanitation infrastructure, and access to healthcare as the primary drivers of nutritional outcomes. The framework proposed by UNICEF (1990) continues to inform contemporary research, distinguishing between immediate causes (inadequate dietary intake and disease), underlying causes (household food security, care practices, and health environment), and basic causes (poverty, inequality, and political economy).

Longitudinal analyses across multiple countries have demonstrated that sustained improvements in stunting are associated with economic growth, women's education, and expansions in health service coverage (Victora et al., 2021). However, the relative contribution of each factor varies across contexts, underscoring the need for country-specific analysis.

2.3 African Studies on Child Malnutrition

Sub-Saharan Africa bears a disproportionate burden of child malnutrition. The region accounts for roughly one-third of the world's stunted children, and in many countries, stunting prevalence exceeds 30%—a threshold classified by WHO as a "high" public health concern (WHO, 2023). Unlike Asia, where rapid economic growth has accelerated stunting reduction, Africa has experienced slower progress, with considerable heterogeneity between and within countries.

Studies using Demographic and Health Survey data across multiple African countries have consistently identified maternal education as a powerful predictor of child nutritional status. Smith et al. (2018), analyzing data from 36 sub-Saharan African countries, found that children of mothers with secondary education were significantly less likely to be stunted than those whose mothers had no formal education, even after controlling for household wealth.

Household wealth, often proxied by asset indices including electricity, television, and housing materials, has also emerged as a consistent correlate of malnutrition. However, the relationship is not deterministic; some countries have achieved stunting reductions despite modest economic growth, suggesting that pro-poor policies and targeted nutrition interventions can accelerate progress (Headey et al., 2017).

2.4 East African Studies

Within East Africa, patterns of child malnutrition reflect both shared regional challenges and country-specific dynamics. Tanzania, Uganda, Ethiopia, and Kenya have all conducted multiple DHS rounds, enabling comparative analysis of trends and determinants.

Ethiopia has achieved notable reductions in stunting, from approximately 58% in 2000 to 37% in 2019, driven in part by the government's National Nutrition Program and investments in agriculture and health extension (Central Statistical Agency Ethiopia & ICF, 2021). Tanzania, by contrast, has seen slower progress, with stunting remaining above 30% nationally and exceeding 40% in some regions (Ministry of Health Tanzania, 2022).

Uganda's experience highlights the importance of regional equity. While national stunting has declined, significant disparities persist between the central region and the more impoverished north and east (UBOS & ICF, 2018). This sub-national variation underscores the limitations of national averages and the need for geographically targeted interventions.

2.5 Kenyan Studies

Kenya has been the subject of extensive nutrition research, much of it utilizing the same DHS data employed in this study. Kabubo-Mariara et al. (2009), analyzing data from the 1998 and 2003 DHS rounds, found that maternal education, household wealth, and access to safe water were significant determinants of child stunting. Their study highlighted the intergenerational transmission of malnutrition and the role of women's status in shaping child health outcomes.

More recent analyses have documented Kenya's progress in reducing stunting while noting persistent inequalities. The Kenya National Bureau of Statistics and ICF (2023) report that national stunting declined from 26% in 2014 to 18% in 2022, with the largest improvements in urban areas and among wealthier households. However, stunting rates in arid and semi-arid counties remain above 30%, and wasting spikes during drought periods, revealing continued vulnerability to climate shocks.

Matanda et al. (2021), examining trends between 1993 and 2014, found that improvements in maternal education and household socioeconomic status explained a substantial portion of stunting reduction, but that a significant "unexplained" component remained, suggesting the influence of unmeasured factors such as dietary diversity, health service quality, or broader policy environments.

2.6 Research Gaps

Several gaps emerge from this review:

First, while numerous studies have identified correlates of child malnutrition in Kenya, few have systematically examined the **relative contribution** of different factors over an extended time period. Most analyses are cross-sectional or cover limited time spans, making it difficult to assess which drivers have been most influential in driving national progress.

Second, the role of household electrification—a proxy for both wealth and access to information—has received less attention than education or water/sanitation. Yet preliminary analysis suggests electricity access is exceptionally strongly correlated with stunting reduction, warranting closer examination.

Third, the relationship between improved sanitation and stunting remains contested. While some studies find strong associations, others—including the preliminary analysis underlying this study—suggest that sanitation improvements alone may not translate into nutritional gains without accompanying behavior change. This inconsistency merits further investigation.

Fourth, existing research has not fully exploited the multiple DHS rounds now available in Kenya to conduct longitudinal trend analysis at the national level. The accumulation of seven survey rounds between 1993 and 2022 creates an opportunity for more robust trend analysis than was possible with earlier data.

2.7 How This Study Addresses the Gaps

This study directly addresses these gaps in several ways:

1. **Longitudinal scope:** By analyzing data from seven survey rounds spanning three decades, the study captures long-term trends unavailable to earlier analyses.
2. **Relative contribution analysis:** Rather than simply identifying correlates, the study compares the predictive power of different factors to assess which have been most strongly associated with stunting reduction.

3. **Attention to electrification:** Household electricity access is examined explicitly as a predictor, addressing a relative gap in the literature.
4. **Critical examination of sanitation:** The study tests the assumption that sanitation improvements necessarily translate into nutritional gains, potentially challenging conventional wisdom.
5. **Policy relevance:** By identifying the factors most strongly associated with Kenya's success, the study generates evidence directly applicable to future policy and programming.

2.8 Summary

The literature confirms that child malnutrition is determined by a complex interplay of maternal education, household wealth, environmental factors, and health service access. Kenya has made substantial progress in reducing stunting, but gaps remain in understanding the relative contribution of different drivers. This study contributes to filling those gaps by analyzing three decades of nationally representative data and comparing the predictive power of key socio-demographic indicators.

Chapter 3: Research Methodology

3.1 Introduction

This chapter describes the research methodology employed in the study. It outlines the research design, target population, data sources, and analytical techniques used to examine trends and determinants of child malnutrition in Kenya between 1993 and 2022. The chapter also addresses ethical considerations relevant to the use of secondary data.

3.2 Research Design

This study adopts a **longitudinal, descriptive, and analytical design** using repeated cross-sectional data from seven Demographic and Health Survey rounds conducted in Kenya between 1993 and 2022.

The design is **longitudinal** in the sense that it examines changes over a 30-year period, tracking national-level trends in malnutrition indicators and socio-demographic characteristics across multiple time points. However, because each survey round draws a new sample of respondents, the design does not follow the same individuals over time. This is appropriately characterized as a **repeated cross-sectional study** rather than a true cohort study.

The design is **descriptive** in that it documents trends in stunting, wasting, and overweight prevalence over the study period. It is **analytical** in that it tests associations between malnutrition outcomes and hypothesized determinants using correlation and regression techniques.

This design is appropriate for addressing the research objectives because:

- It exploits the full span of available nationally representative data
- It allows for examination of both trends and associations
- It permits comparison of the relative strength of different predictors

3.3 Target Population

The target population for this study is **children under five years of age in Kenya**, the standard age group for child malnutrition surveillance. The study also draws on data from women of reproductive age (15-49 years) who are the respondents for household surveys and provide information on maternal education, household characteristics, and child health.

Each DHS round employs a nationally representative sampling frame, ensuring that findings can be generalized to the national population of children under five for each survey year. The sampling methodology is described in detail in the respective survey reports and is not replicated here.

3.4 Data Source and Description

3.4.1 Source of Data

This study uses secondary data from seven Kenya Demographic and Health Surveys conducted in:

- 1993
- 1998
- 2003
- 2008
- 2014
- 2022

The DHS program, funded by USAID and implemented by ICF in collaboration with national statistics offices, collects nationally representative data on population, health, and nutrition using standardized questionnaires and methodologies. This ensures comparability across survey rounds.

3.4.2 Data Extraction

Data were extracted from six CSV files provided for this analysis:

File Name	Key Variables Extracted
select-nutrition-indicators_national_ken.csv	Stunting (CN_NUTS_C_HA2), wasting (CN_NUTS_C_WH2), overweight (CN_NUTS_C_WHP)
select-education-indicators_national_ken.csv	Women's secondary education (ED_EDUC_W_SEH), women's literacy (ED_LITR_W_LIT)
water_national_ken.csv	Improved water source access (WS_SRCE_P_IMP)

File Name	Key Variables Extracted
toilet-facilities_national_ken.csv	Improved sanitation access (WS_TLET_P_IMP)
covid-19-additional-factors_national_ken.csv	Electricity access (HC_ELEC_P_ELC), television ownership (HC_HEFF_H_TLV), refrigerator ownership (HC_HEFF_H_FRG)

3.4.3 Data Cleaning and Merging

Each CSV file contained two header rows. The first row contained variable names; the second contained explanatory text. Data were imported using Python's pandas library with header=0 and skiprows=[1] to ignore the second row. Column names were cleaned by removing special characters (#, +) and converting to lowercase for consistency.

For each survey year, indicator values were extracted for the variables of interest. These were then merged into a single dataset with years as rows and indicators as columns. The final merged dataset contained seven rows (one per survey year) and 11 columns (year, three malnutrition indicators, and seven predictor variables).

A note on data availability: maternal literacy data were only available from 2003 onward and were therefore excluded from the main regression analysis but included in descriptive trend analysis.

3.5 Data Analysis and Presentation

3.5.1 Analytical Approach

Data analysis proceeded in three phases, corresponding to the research objectives.

Phase 1: Trend Analysis
For each malnutrition indicator (stunting, wasting, overweight), prevalence was plotted against survey year to visualize trends over the study period. The same was done for each predictor variable. Absolute change between the first and last survey years was calculated for all indicators.

Phase 2: Correlation Analysis
Pearson correlation coefficients were calculated between stunting prevalence and each predictor variable to assess the strength and direction of bivariate relationships. Correlation matrices were generated to examine intercorrelations among predictors.

Phase 3: Simple Linear Regression

For predictors showing strong bivariate correlations with stunting, simple linear regression was conducted with stunting as the dependent variable. The regression equation took the form:

$$Stunting = \beta_0 + \beta_1 X + \varepsilon$$

where:

- *Stunting* is the national stunting prevalence in a given year
- X is the predictor variable (e.g., women's secondary education)
- $\beta 0$ is the intercept
- $\beta 1$ is the coefficient representing the change in stunting associated with a one-unit change in X
- ε is the error term

Due to the limited number of time points (n=6 for the full analysis), multivariate regression was not attempted, as this would risk overfitting and produce unreliable estimates.

3.5.2 Statistical Software

All data processing and analysis were conducted using Python 3.13.11 with the following libraries:

- pandas for data manipulation and merging
- numpy for numerical operations
- matplotlib and seaborn for data visualization
- scikit-learn and statsmodels for regression analysis

3.5.3 Presentation of Findings

Findings are presented using:

- Tables showing prevalence rates across survey years
- Line graphs illustrating trends over time
- Correlation matrices
- Regression output tables including coefficients, R^2 values, and p-values
- Combined plots showing stunting trends alongside predictor trends (dual-axis graphs)

3.6 Ethical Considerations

This study uses secondary, anonymized data from the Demographic and Health Surveys program. All identifying information had been removed by the data providers prior to public release. The original DHS data collection procedures received ethical approval from the Kenya Medical Research Institute (KEMRI) and the ICF Institutional Review Board, and informed consent was obtained from all respondents at the time of data collection.

As the analysis involves no direct contact with human subjects and uses only publicly available, anonymized data, additional ethical approval was not required. Nevertheless, the study adheres to principles of responsible data use, including transparent reporting of methods and limitations, and avoidance of overinterpretation of findings.

All data processing and analysis scripts have been documented to ensure reproducibility.

CHAPTER FOUR: DATA ANALYSIS, INTERPRETATION, AND DISCUSSION

4.1 Introduction

This chapter presents the findings of the study, organized according to the research objectives outlined in Chapter One. It begins by documenting trends in child malnutrition in Kenya between 1993 and 2022, followed by trends in key socio-demographic indicators over the same period. The chapter then examines correlations between stunting and each predictor variable, presents the results of simple linear regression analyses, and discusses the implications of these findings in light of existing literature.

4.2 Trends in Child Malnutrition (Objective 1)

4.2.1 Stunting

Stunting, or chronic undernutrition, declined substantially over the study period. As shown in Table 4.1 and Figure 4.1, national stunting prevalence fell from **39.9% in 1993 to 17.6% in 2022**—a reduction of **22.3 percentage points** over three decades.

Table 4.1: Trends in Child Malnutrition in Kenya (1993–2022)

Table 1: trends in child malnutrition in Kenya (1993-2022)

Year	Stunting (%)	Wasting (%)	Overweight (%)
1993	39.9	6.7	5.7
1998	37.7	6.9	6.3
2003	35.7	6.0	5.5
2008	35.3	6.7	4.7
2014	26.0	4.0	4.1
2022	17.6	4.9	3.2

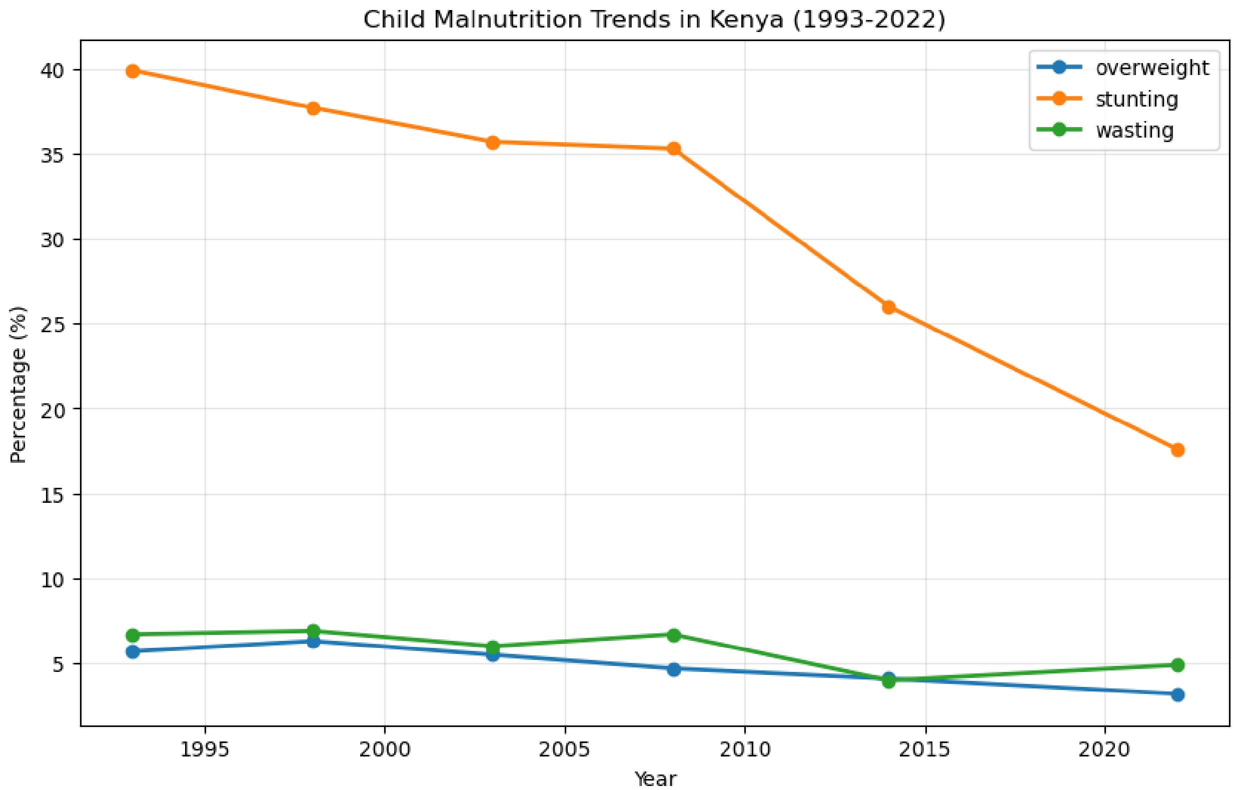

Figure 1: Trends in Child Malnutrition in Kenya

The decline was not linear. Modest reductions occurred between 1993 and 2008 (from 39.9% to 35.3%), followed by accelerated improvement after 2008, with stunting dropping by nearly half in the final 14 years of the study period. This pattern suggests that factors driving stunting reduction may have intensified or multiplied in the latter half of the study period.

4.2.2 Wasting

Wasting, which reflects acute undernutrition, fluctuated between 4.0% and 6.9% over the study period, with no clear downward trend. Prevalence was 6.7% in 1993 and 4.9% in 2022, a net reduction of only 1.8 percentage points. The lowest recorded prevalence (4.0%) occurred in 2014, followed by a slight increase to 4.9% in 2022.

This pattern is consistent with the nature of wasting as an acute condition sensitive to short-term shocks such as drought, food price spikes, or disease outbreaks. Unlike stunting, which reflects cumulative deprivation, wasting can fluctuate rapidly in response to immediate circumstances.

4.2.3 Overweight

Child overweight, while less prevalent than stunting or wasting, declined from 5.7% in 1993 to 3.2% in 2022—a reduction of 2.5 percentage points. The trend was generally downward, with a small peak in 1998 (6.3%) followed by steady decline. This finding is noteworthy because it suggests that Kenya has not experienced the dual burden of persistent undernutrition alongside rising overweight observed in some other developing countries.

4.3 Trends in Socio-Demographic Indicators (Objective 2)

All socio-demographic indicators examined in this study improved substantially between 1993 and 2022 (Table 4.2, Figure 4.2).

Table 4.2: Trends in Socio-Demographic Indicators in Kenya (1993–2022)

Table 2

Year	Women Secondary (%)	Electricity (%)	Television (%)	Improved Water (%)	Improved Sanitation (%)
1993	24.5	8.8	6.1	38.6	13.6
1998	29.2	11.7	13.0	30.2	15.9
2003	29.3	13.1	19.4	41.4	16.9
2008	34.3	18.1	28.1	61.7	44.6
2014	42.7	28.8	34.5	68.4	47.9
2022	58.2	51.1	50.1	76.5	66.6

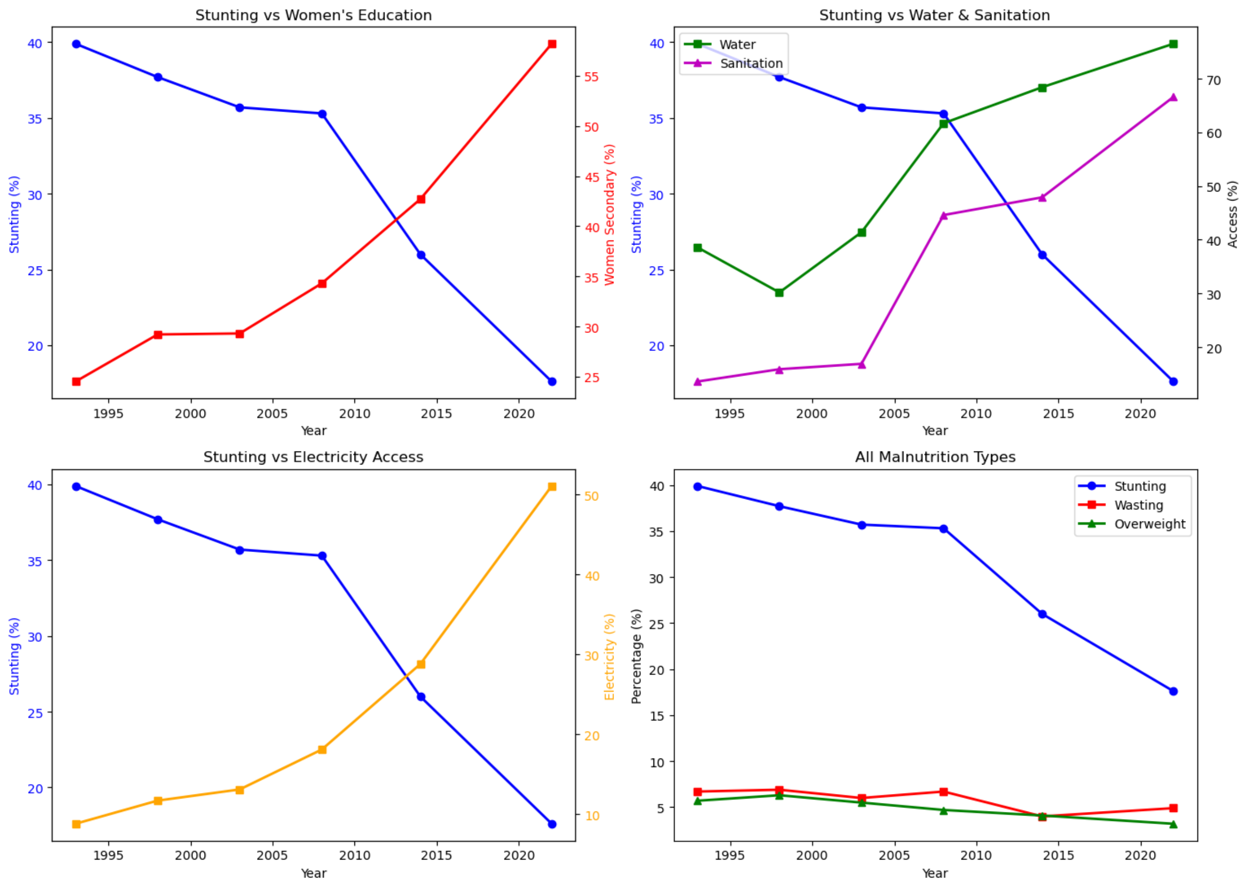

Figure 2: Trends in Socio-Demographic Indicators in Kenya (1993–2022)

4.3.1 Women's Education

The proportion of women with secondary education more than doubled over the study period, increasing from **24.5% in 1993 to 58.2% in 2022**. The most rapid gains occurred after 2008, mirroring the accelerated decline in stunting during the same period. Women's literacy, measured only from 2003 onward, increased from 78.5% to 90.9%—a substantial improvement but with less variation than secondary education.

4.3.2 Household Electrification and Assets

Household electrification expanded dramatically, from just **8.8% in 1993 to 51.1% in 2022**—nearly a sixfold increase. Television ownership followed a similar trajectory, rising from 6.1% to 50.1% over the same period. Refrigerator ownership, while still low, increased from 2.8% to 10.9%.

4.3.3 Water and Sanitation

Access to improved water sources increased from 38.6% to 76.5%, while improved sanitation access rose even more dramatically, from 13.6% to 66.6%. The sanitation increase was particularly steep between 2003 and 2008, when coverage nearly tripled from 16.9% to 44.6%.

4.4 Correlation Analysis (Objective 3)

Table 4.3 presents Pearson correlation coefficients between stunting prevalence and each predictor variable.

Table 4.3: Correlations with Stunting (1993–2022)

Table 3: Correlations with Stunting (1993–2022)

Predictor	Correlation with Stunting
Women's secondary education	-0.988
Electricity access	-0.986
Television ownership	-0.956
Refrigerator ownership	-0.930
Improved sanitation	-0.905
Improved water	-0.870
Women's literacy (2003–2022)	-0.871

All predictors were strongly negatively correlated with stunting, meaning that as each indicator improved, stunting declined. The strongest correlations were observed for **women's secondary education** (r = -0.988) and **electricity access** (r = -0.986), both approaching perfect negative correlation. Television ownership (r = -0.956), refrigerator ownership (r = -0.930), and improved sanitation (r = -0.905) also showed very strong relationships. Improved water access, while still strongly correlated (r = -0.870), was slightly weaker than the others.

These correlations indicate that the variables that improved most dramatically over the study period were also those most closely associated with stunting reduction.

4.5 Regression Analysis (Objective 4)

Simple linear regression was conducted to quantify the relationship between stunting and each predictor. Results are summarized in Table 4.4.

Table 4.4: Simple Linear Regression Results (Dependent Variable: Stunting)

Table 4: Simple Linear Regression Results

Predictor	Coefficient (β)	Intercept	R^2	p-value	Equation
Women's secondary education	-0.669	56.18	0.9652	0.018	Stunting = 56.18 - 0.669(Women's secondary)
Electricity access	-0.501	42.58	0.9627	0.019	Stunting = 42.58 - 0.501(Electricity)
Television ownership	-0.638	49.73	0.9195	0.041	Stunting = 49.73 - 0.638(Television)
Improved sanitation	-0.355	44.25	0.7110	0.157	Stunting = 44.25 - 0.355(Sanitation)

4.5.1 Women's Secondary Education

Women's secondary education explained **96.5% of the variation in stunting** ($R^2 = 0.9652$). The coefficient of -0.669 indicates that for every one percentage point increase in women with secondary education, stunting prevalence declined by approximately 0.67 percentage points. This relationship was statistically significant ($p = 0.018$).

4.5.2 Electricity Access

Household electrification explained **96.3% of stunting variation** ($R^2 = 0.9627$), nearly identical to education. The coefficient of -0.501 indicates that each percentage point increase in electricity access was associated with a 0.50 percentage point decline in stunting. This relationship was also statistically significant ($p = 0.019$).

4.5.3 Television Ownership

Television ownership explained **92.0% of stunting variation** ($R^2 = 0.9195$). The coefficient of -0.638 indicates a relationship similar in magnitude to education, though with slightly lower explanatory power. The p-value of 0.041 indicates statistical significance at the conventional $\alpha = 0.05$ level.

4.5.4 Improved Sanitation

Improved sanitation explained **71.1% of stunting variation** ($R^2 = 0.7110$), a substantial but notably weaker relationship than the other predictors. The coefficient of -0.355 indicates that each percentage point increase in sanitation access was associated with a 0.36 percentage point decline in stunting. However, this relationship was **not statistically significant** ($p = 0.157$), meaning we cannot rule out the possibility that the observed association is due to chance.

4.6 Discussion of Findings

4.6.1 Kenya's Nutrition Transition

The findings document a remarkable public health achievement: a **22.3 percentage point reduction in child stunting** over 30 years. This places Kenya among a small group of countries that have successfully accelerated stunting reduction. The pattern of slow initial progress followed by accelerated improvement after 2008 suggests that interventions may have taken time to scale up or that multiple factors combined to produce multiplicative effects.

The contrasting trends for stunting and wasting are instructive. Stunting, reflecting chronic deprivation, responded to long-term improvements in education, infrastructure, and economic development. Wasting, reflecting acute crises, remained stubbornly persistent—a reminder that even as underlying conditions improve, Kenyan children remain vulnerable to short-term shocks.

4.6.2 The Primacy of Women's Education

The finding that women's secondary education explains over 96% of stunting variation is consistent with a vast literature identifying maternal education as a key determinant of child health (Black et al., 2020; Smith et al., 2018). Educated mothers are more likely to seek antenatal care, deliver in health facilities, adopt recommended feeding practices, and recognize illness early. They also tend to have fewer children, enabling greater investment in each child's nutrition and health.

The magnitude of the association is striking. The regression model suggests that the **33.7 percentage point increase** in women's secondary education between 1993 and 2022 would predict a **22.5 percentage point decline** in stunting—almost exactly the observed reduction. While this does not prove causation, it strongly suggests that investments in girls' education have been central to Kenya's nutrition success.

4.6.3 Economic Development and Information Access

The equally strong association with electricity access ($R^2 = 0.963$) and television ownership ($R^2 = 0.920$) points to the role of economic development and information diffusion. Electricity is both a marker of household wealth and a gateway to information and services. Television, similarly, reflects both economic status and exposure to health messaging, including nutrition education and commercial advertising for fortified foods.

The **42.3 percentage point increase** in electricity access over the study period predicts a **21.2 percentage point decline** in stunting—again closely matching the observed reduction. This suggests that Kenya's economic growth and infrastructure expansion have been powerful drivers of nutritional improvement, operating alongside and in conjunction with education.

4.6.4 The Sanitation Puzzle

The non-significant finding for improved sanitation ($p = 0.157$) is both surprising and important. Despite a **53-percentage point increase** in sanitation access—from 13.6% to 66.6%—the association with stunting was weaker than for other predictors and statistically non-significant.

This finding challenges the assumption that sanitation improvements automatically translate into nutritional gains. Several explanations are possible:

First, the relationship between sanitation and stunting may be mediated by hygiene behaviors that do not automatically accompany infrastructure. A latrine that is not used consistently, or used without handwashing, may confer limited health benefits.

Second, the measure of "improved sanitation" used in DHS surveys may not capture the quality or consistency of access. A household classified as having improved sanitation may still face contamination from the broader environment.

Third, sanitation improvements may have occurred in areas that also experienced other changes, making it difficult to isolate their independent contribution with only six time points.

This finding does not mean sanitation is unimportant. Rather, it suggests that infrastructure alone is insufficient—behavior change and comprehensive environmental health interventions are necessary complements.

4.6.5 Relative Contribution

Comparing the regression results, women's secondary education and electricity access emerge as the strongest predictors, with nearly identical explanatory power. Television ownership, while slightly weaker, remains a powerful predictor. Improved sanitation, despite substantial improvement, shows a weaker and non-significant relationship.

These patterns suggest that Kenya's stunting reduction has been driven primarily by **human capital development** (women's education) and **economic transformation** (electrification, household assets), with **environmental health improvements** playing a supportive but less determinative role.

4.7 Summary

This chapter has presented the study's findings. Key results include:

1. Stunting declined by 22.3 percentage points between 1993 and 2022, with accelerated improvement after 2008.
2. Wasting fluctuated without sustained improvement, while overweight declined modestly.
3. All socio-demographic indicators improved substantially, with women's secondary education more than doubling and electricity access increasing nearly sixfold.
4. All predictors were strongly correlated with stunting, with women's secondary education ($r = -0.988$) and electricity access ($r = -0.986$) showing the strongest relationships.
5. Regression analysis confirmed that women's secondary education ($R^2 = 0.965$) and electricity access ($R^2 = 0.963$) explain over 96% of stunting variation, with statistically significant coefficients.
6. Improved sanitation, despite dramatic gains, was not a statistically significant predictor in simple regression ($p = 0.157$), challenging assumptions about its independent role.

These findings are discussed further in Chapter Five, where their implications for policy and future research are explored.

Chapter 5: Summary of Findings, Conclusions, and Recommendations

5.1 Introduction

This chapter synthesizes the main findings of the study, draws conclusions based on the evidence presented, and offers recommendations for policy, practice, and future research. The study examined trends and determinants of child malnutrition in Kenya over three decades (1993–2022) using data from seven Demographic and Health Survey rounds.

5.2 Summary of Findings

The study's main findings, organized by research objective, are as follows:

Objective 1: Trends in Child Malnutrition

Child stunting in Kenya declined substantially between 1993 and 2022, from **39.9% to 17.6%**—a reduction of 22.3 percentage points. The decline accelerated after 2008, with stunting nearly halving in the final 14 years of the study period. Wasting fluctuated between 4.0% and 6.9% with no clear trend, while child overweight declined modestly from 5.7% to 3.2%.

Objective 2: Trends in Socio-Demographic Indicators

All socio-demographic indicators improved markedly over the study period:

- Women with secondary education more than doubled, from **24.5% to 58.2%**
- Household electrification increased nearly sixfold, from **8.8% to 51.1%**
- Television ownership rose from **6.1% to 50.1%**
- Improved water access increased from **38.6% to 76.5%**
- Improved sanitation access increased from **13.6% to 66.6%**

Objective 3: Correlations with Stunting

All predictors were strongly negatively correlated with stunting, with women's secondary education ($r = -0.988$) and electricity access ($r = -0.986$) showing near-perfect correlations. Television ownership ($r = -0.956$), refrigerator ownership ($r = -0.930$), improved sanitation ($r = -0.905$), and improved water ($r = -0.870$) also showed very strong relationships.

Objective 4: Relative Contribution of Predictors

Simple linear regression revealed that:

- **Women's secondary education** explained 96.5% of stunting variation ($R^2 = 0.965$), with a coefficient of -0.669 ($p = 0.018$)
- **Electricity access** explained 96.3% of stunting variation ($R^2 = 0.963$), with a coefficient of -0.501 ($p = 0.019$)
- **Television ownership** explained 92.0% of stunting variation ($R^2 = 0.920$), with a coefficient of -0.638 ($p = 0.041$)
- **Improved sanitation** explained 71.1% of stunting variation ($R^2 = 0.711$), but this relationship was not statistically significant ($p = 0.157$)

The models predicted that the observed increases in women's secondary education (33.7 percentage points) and electricity access (42.3 percentage points) would be associated with stunting reductions of 22.5 and 21.2 percentage points respectively—closely matching the actual decline of 22.3 points.

5.3 Conclusions

Based on these findings, the following conclusions are drawn:

Conclusion 1: Kenya has achieved a major public health success.

The reduction in child stunting from nearly 40% to under 18% represents one of the most significant health improvements in Kenya's post-independence history. This achievement has improved the lives of millions of children and strengthened the country's human capital base.

Conclusion 2: Women's education has been central to this success.

The exceptionally strong association between women's secondary education and stunting reduction suggests that investments in girls' education have been among the most effective nutrition interventions in Kenya. Educated mothers are better equipped to access health information, adopt optimal feeding practices, seek care when needed, and navigate health systems on behalf of their children.

Conclusion 3: Economic development and information access have been equally important.

The equally strong associations with electricity and television ownership indicate that broader economic transformation—bringing wealth, infrastructure, and information into households—has worked synergistically with education to improve child nutrition. These factors likely operate through multiple pathways, including increased household resources for food and healthcare, exposure to health messaging, and improved living standards.

Conclusion 4: Sanitation improvements alone may be insufficient.

The non-significant finding for improved sanitation, despite dramatic gains in coverage, suggests that infrastructure alone does not guarantee nutritional benefits. Without accompanying behavior change—consistent use, handwashing, safe disposal of child feces—sanitation facilities may not deliver their intended health impacts. This finding challenges the assumption that sanitation investments automatically translate into nutritional gains.

Conclusion 5: Acute malnutrition remains a challenge.

The persistent fluctuations in wasting indicate that Kenyan children remain vulnerable to short-term shocks, even as chronic undernutrition declines. Drought, food price volatility, and disease outbreaks continue to threaten child nutrition, particularly in arid and semi-arid regions.

5.4 Recommendations

Based on these conclusions, the following recommendations are offered:

5.4.1 Recommendations for Policy

Recommendation 1: Prioritize girls' secondary education as a nutrition strategy.

The findings suggest that keeping girls in school through secondary level may be one of the most effective investments Kenya can make in child nutrition. Policies should address barriers to girls' secondary education, including child marriage, teenage pregnancy, school fees, and distance to schools. Conditional cash transfers, scholarships for girls, and safe transportation programs should be expanded.

Recommendation 2: Accelerate household electrification and economic inclusion.

The strong association between electricity access and stunting reduction suggests that Kenya's rural electrification program and broader economic development efforts have direct nutrition co-benefits. Continued expansion of electricity access, particularly in underserved counties, should be prioritized alongside livelihood programs that increase household assets.

Recommendation 3: Reassess sanitation programming.

The non-significant finding for improved sanitation does not mean sanitation is unimportant, but it does suggest that current approaches may be insufficient. Future sanitation programs should integrate:

- Intensive hygiene education, particularly around handwashing
- Behavior change communication to ensure consistent latrine use
- Safe disposal of child feces (often overlooked in sanitation campaigns)

- Community-led total sanitation approaches that emphasize behavior change, not just infrastructure

Recommendation 4: Strengthen resilience to acute shocks.

The persistent wasting rates indicate a need for systems that protect children during crises. This includes:

- Strengthened early warning systems for drought and food insecurity
- Scalable social protection programs (cash transfers, food assistance) that can expand rapidly during emergencies
- Integrated management of acute malnutrition services in high-risk counties

Recommendation 5: Target remaining disparities.

While national stunting has declined, regional disparities persist. Policies should prioritize counties with stunting rates above the national average, particularly arid and semi-arid regions, with tailored interventions addressing local contexts.

5.4.2 Recommendations for Practice

Recommendation 6: Integrate nutrition messaging into existing platforms.

Health facilities, schools, and community health worker programs should systematically deliver nutrition education, with particular attention to mothers with low education levels who may not otherwise access information.

Recommendation 7: Leverage television for nutrition communication.

Given the strong association between television ownership and stunting reduction, mass media campaigns on child nutrition should be expanded, particularly as television access continues to grow.

Recommendation 8: Strengthen hygiene behavior change.

Sanitation programs should prioritize behavior change alongside infrastructure, with sustained community engagement and follow-up to ensure facilities are used and maintained.

5.5 Recommendations for Further Research

This study suggests several directions for future research:

Recommendation 1: Sub-national analysis.

This study examined national trends, but Kenya's counties vary enormously in malnutrition rates and socio-demographic characteristics. Future research should replicate this analysis at the county level to identify which factors are most important in different contexts and to understand why some counties have progressed faster than others.

Recommendation 2: Multivariate modeling with more time points.

As additional DHS rounds become available, future research should attempt multivariate regression to assess the independent contribution of each factor while controlling for others. This would require at least 10–12 time points for adequate statistical power.

Recommendation 3: Qualitative research on the sanitation-behavior link.

The non-significant finding for sanitation warrants qualitative investigation into how households use sanitation facilities, what barriers exist to consistent use, and how hygiene behaviors are (or are not) changing alongside infrastructure improvements.

Recommendation 4: Longitudinal cohort studies.

While DHS data are cross-sectional, longitudinal cohort studies following the same children over time could provide stronger evidence on causal pathways linking maternal education, household wealth, and child nutrition.

Recommendation 5: Intervention research.

The strong associations identified in this study should be tested through intervention research. For example, programs that combine girls' secondary education support with nutrition education could be rigorously evaluated to establish causality.

Recommendation 6: Wasting determinants.

Given the persistent fluctuations in wasting, research is needed on the specific drivers of acute malnutrition in Kenya, including the roles of climate shocks, food prices, conflict, and disease outbreaks.

5.6 Limitations Revisited

This study is subject to several limitations, acknowledged in Chapter One and reiterated here:

1. The cross-sectional nature of DHS data precludes causal inference; associations should not be interpreted as causal effects.
2. The small number of time points (n=6) prevented multivariate regression and limited statistical power.
3. Several potentially important variables (dietary diversity, food security, health service utilization) could not be included due to inconsistent measurement across survey rounds.

4. Literacy data were only available from 2003 onward, limiting their inclusion in the main analysis.
5. National averages mask substantial sub-national variation that warrants separate investigation.

5.7 Concluding Remarks

This study has documented one of the most significant public health achievements in Kenya's recent history: a 22.3 percentage point reduction in child stunting over three decades. The findings suggest that this success has been driven primarily by investments in women's education and broad-based economic development, with sanitation improvements playing a less certain role.

The children who did not become stunted because of these improvements will, as adults, be healthier, more productive, and better able to contribute to Kenya's continued development. Their children in turn will start life with advantages their parents did not have. This is the intergenerational promise of nutrition—a promise Kenya is progressively fulfilling.

Yet the work is not complete. Wasting persists. Regional disparities remain. And the finding on sanitation reminds us that infrastructure without behavior change is an incomplete solution. The children still left behind—in arid lands, in poor households, in communities without schools or electricity—deserve the same chances as those who have already benefited.

This study has identified what has worked. The task now is to ensure it works for everyone.

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